



Lesson 9

Land Use and Management





Objectives

1. Identify and describe the different types of land use.
2. Explain the importance of land.
3. Analyze issues in land use and suggest possible solutions.
4. Discuss sustainable land management practices.
5. Explain RA 9003 and its role in managing waste and protecting the environment.

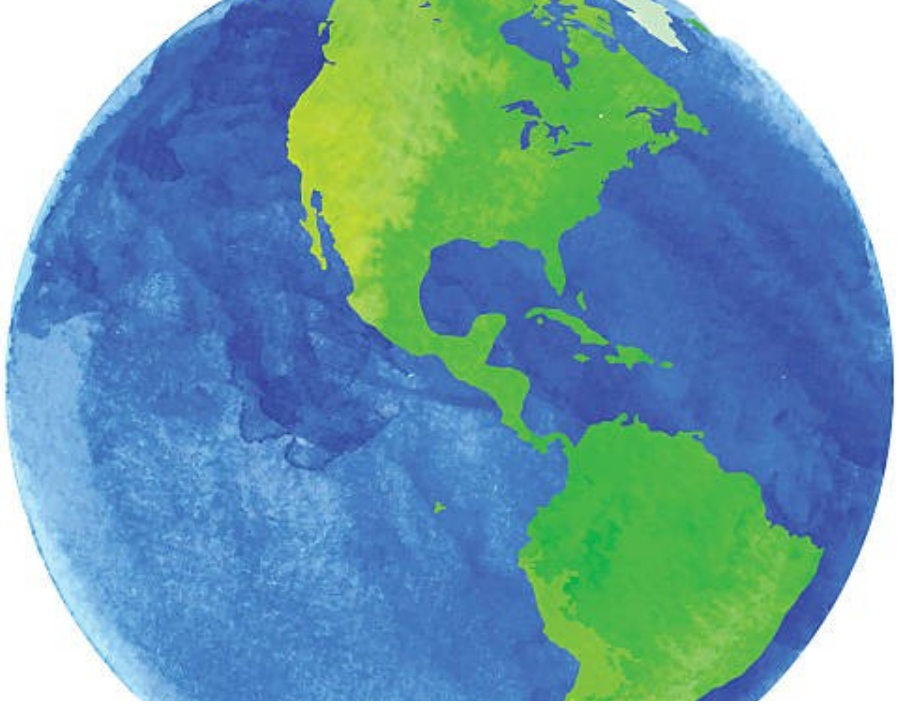
Have you ever wondered why some places are filled with farms and trees, while others are full of buildings and roads?





The Land

- Land is the solid, dry surface of the Earth, making up about 29.2% of the planet's surface.
- It includes soil, rocks, and minerals, and is where natural resources are found on and beneath the surface.





Importance of Land

1. **Habitat** – Provides living space for humans, animals, and plants.
2. **Agriculture** – Supplies the soil and nutrients necessary for growing crops and raising livestock.
3. **Natural Resources** – Contains valuable minerals, fossil fuels, and construction materials.
4. **Infrastructure** – Serves as the foundation for homes, roads, industries, and other human structures.
5. **Ecological Balance** – Supports ecosystems and maintains biodiversity.

Land Use

- **Land Use** describes the **purpose or activity** for which a piece of land is utilized.



Types of Land Use

1. **Agricultural Land –** Used for farming, livestock, and other food production.
2. **Residential Land –** Areas for housing and communities.
3. **Commercial Land –** Used for businesses, trade, and services.
4. **Industrial Land –** For factories, manufacturing, and processing plants.
5. **Institutional/Public land** - For municipal facilities, hospitals, museums, religious buildings, and schools.

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6. **Recreational Land –** Parks, playgrounds, sports fields, and tourism sites.
7. **Transport Land** – Roads, railways, airports, and seaports.
8. **Forest Land** – Covered with trees and vegetation that support wildlife and prevent erosion.
9. **Protected Areas –** conservation zones for wildlife and natural habitats.

Issues in Land Use

1. Deforestation
2. Soil erosion and degradation
3. Urban sprawl
4. Land conversion

Deforestation

Deforestation is the large-scale removal of trees and forests for human activities such as logging, agriculture, or urban development.

- **Effects**
  - Loss of habitat for wildlife.
  - Increased carbon dioxide in the atmosphere (since fewer trees absorb CO_2).
  - Disruption of the water cycle.
  - Contribution to soil erosion and climate change.



Soil Erosion and Degradation

Soil erosion is the wearing away of the top layer of soil due **to wind, water,** or **human activities** such as **over-farming** and **deforestation.**

- **Effects**
  - Loss of fertile soil, reducing crop productivity.
  - Sedimentation in rivers and lakes, leading to flooding.
  - Difficulty in plant growth due to loss of nutrients.
  - Desertification in extreme cases.



Urban Sprawl

Urban sprawl refers to the **spread of cities** into the **countryside or rural areas**.

This occurs when more houses, roads, malls, and buildings are constructed outside the city, causing cities to expand and cover a larger area of **land**.



Effects

- Loss of agricultural and forest lands.
- Increased traffic and pollution.
- Higher demand for water, energy, and other resources.
- Demands on infrastructure such as roads, schools, and hospitals.

Land Conversion
(from Agricultural to Industrial/Residential Use)

Land conversion happens when farmland is turned into areas for housing, factories, or commercial establishments.

- **Effects**
  - Reduction in available land for food production.
  - Increased risk of flooding due to loss of soil and vegetation.
  - Displacement of farming communities.
  - Greater dependency on imported food supplies.



LAND MANAGEMENT

- **Land Management** refers to the **methods, policies, and practices** to use land resources **wisely and sustainably.**



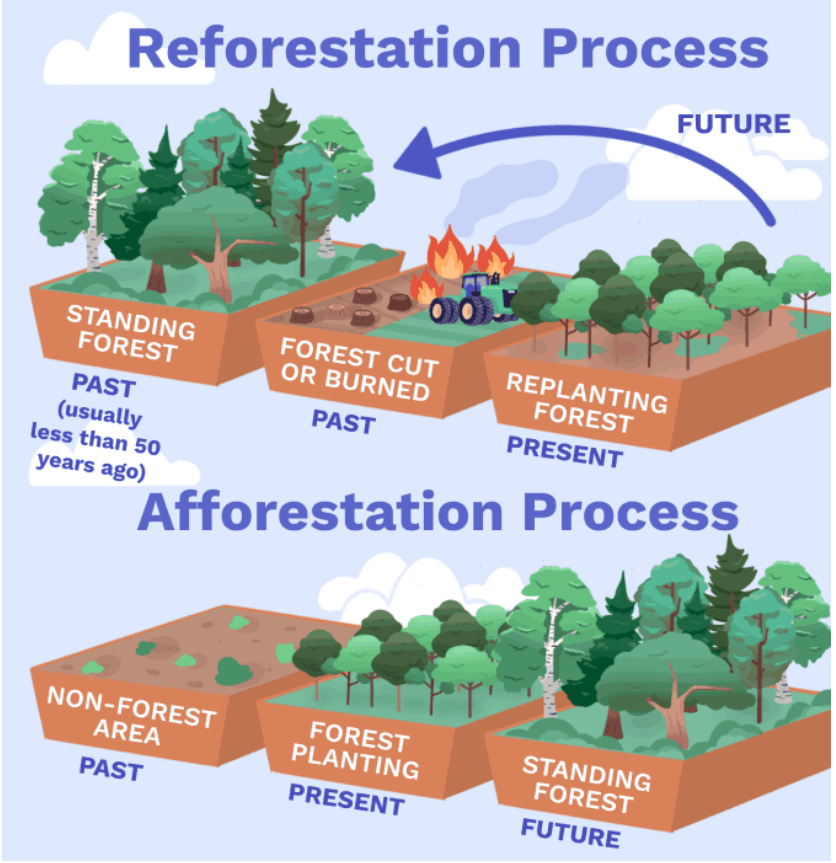
Sustainable Land Management Practices

1. Reforestation and Afforestation
2. Contour Plowing and Terracing
3. Organic and Conservation Farming
4. Agroforestry (trees + crops)
5. Land Zoning and Protected Areas
6. Use of Green Infrastructure
7. Proper Waste Disposal

Reforestation and Afforestation

- **Reforestation** means **replanting trees** in areas where forests were cut down or destroyed.
- **Afforestation** means **planting trees** in areas that were not previously forested.

- **Purpose**
  - Prevents soil erosion
  - Improves air quality
  - Restores habitats for wildlife
  - Helps absorb carbon dioxide and combat climate change



Contour Plowing and Terracing

- **Contour Plowing:** Plowing along the natural curves or contours of the land to slow down water runoff.
- **Terracing:** Creating step-like flat areas on slopes to prevent soil erosion and make hilly land suitable for farming.

Purpose

- Reduces water loss and soil erosion
- Improves water absorption by the soil
- Increases crop productivity on slopes





Organic and Conservation Farming

- **Organic Farming:** Avoids synthetic fertilizers and pesticides, instead using compost, **crop rotation**, and **natural pest control**.
- **Conservation Farming:** Minimizes **soil disturbance**, maintains **crop cover**, and **rotates crops** to keep the soil healthy.

Purpose

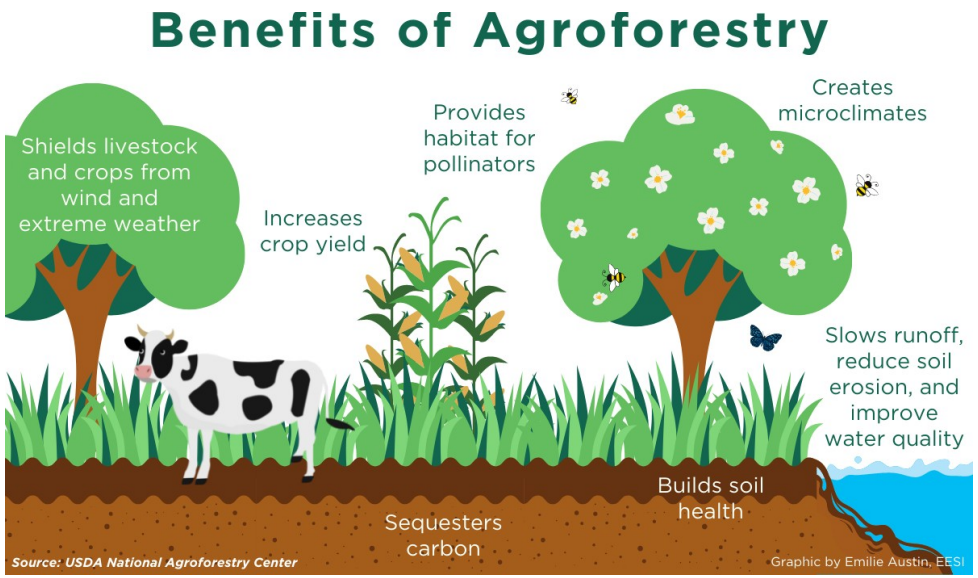
- Maintains soil fertility and biodiversity
- Protects groundwater from chemical pollution
- Promotes long-term agricultural sustainability



Agroforestry (Trees + Crops)

- The practice of **growing trees and crops together** on the same land.

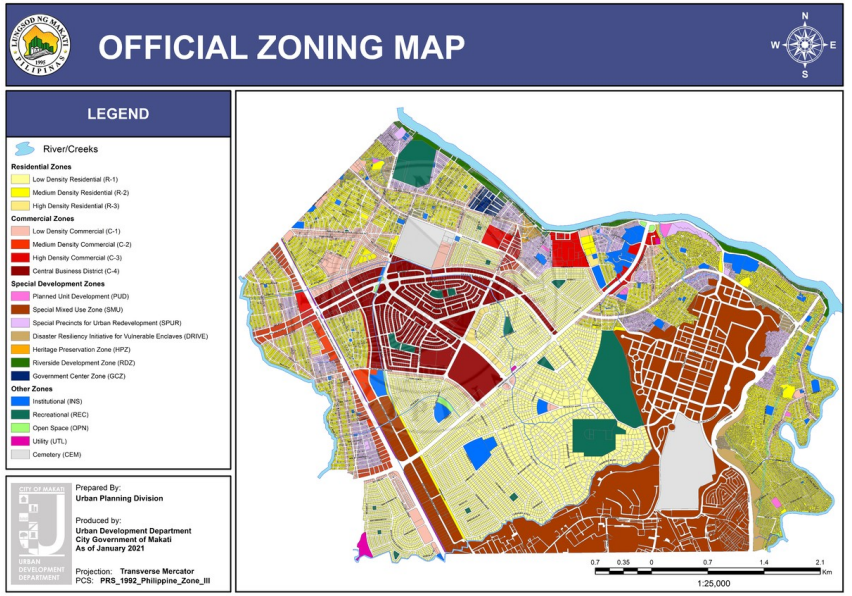
- **Purpose**
  - Prevents erosion and improves soil fertility.
  - Provides shade, fruits, and additional income.
  - Enhances biodiversity and reduces the need for chemical inputs.



Land Zoning and Protected Areas

- **Land Zoning:** Dividing land into zones (residential, agricultural, industrial, etc.) to guide proper land use.
- **Protected Areas:** Declaring forests, watersheds, and ecosystems as protected zones.

- **Purpose**
  - Prevents land-use conflicts and overdevelopment
  - Conserves natural habitats and biodiversity
  - Ensures sustainable land allocation for different needs

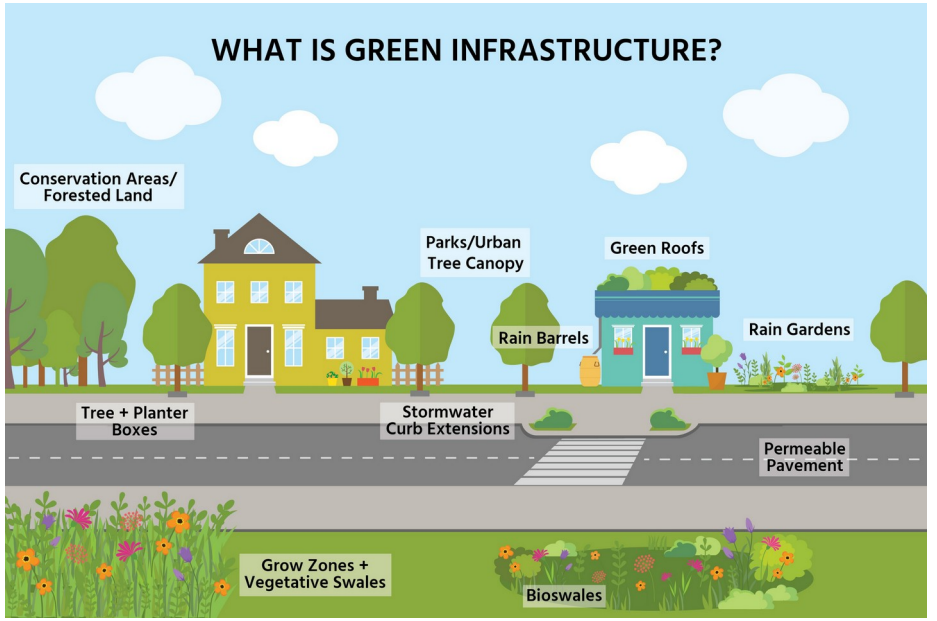




Use of Green Infrastructure

- Designing cities and communities with **nature-based solutions** like **parks, green roofs, buffer zones, and rain gardens**.

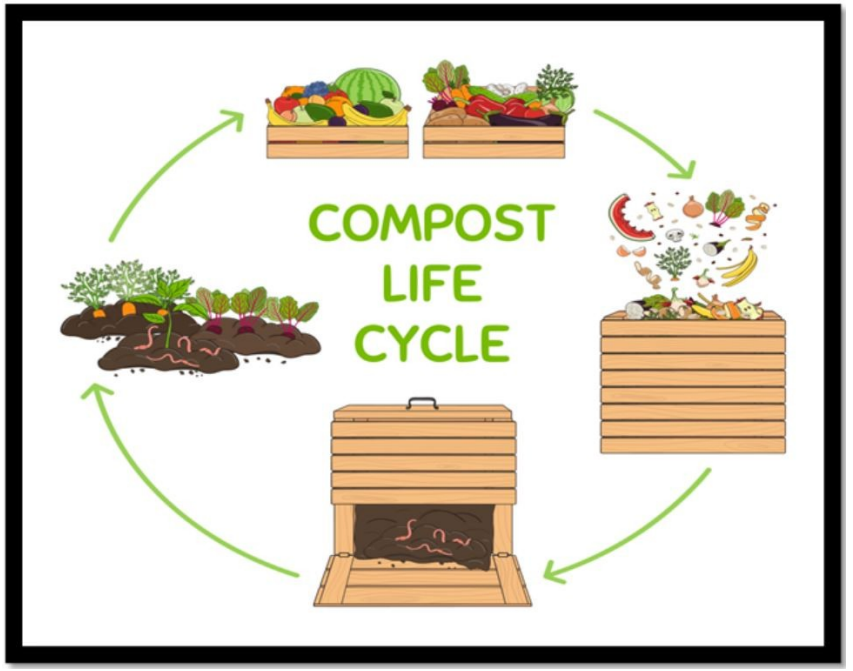
- **Purpose**
  - Reduces flooding and improves air quality
  - Enhances urban biodiversity
  - Provides recreation and improves quality of life



Proper Waste Disposal

- **Proper Waste Disposal** means managing household, agricultural, and industrial waste properly to prevent pollution and land contamination.
  Practices:
  - Waste segregation and recycling
  - Composting of biodegradable materials
  - Safe disposal of hazardous waste

- **Purpose**
  - Keeps land and water sources clean
  - Reduces greenhouse gas emissions from dumpsites





RA 9003

- The **Ecological Solid Waste Management Act of 2000** (Republic Act No. 9003) is a law that **provides a comprehensive framework for proper solid waste management**.
- It emphasizes **reducing, reusing, recycling, and proper disposal of solid waste**.



Key Provisions

1. **Waste Segregation**
   - Households, commercial establishments, and institutions must **segregate waste into biodegradable, non-biodegradable, and recyclable materials**.
2. **Collection and Transport**
   - Local government units (LGUs) are responsible for **collecting and transporting waste** efficiently and sustainably.
3. **Recycling and Composting**
   - Encourages **resource recovery facilities** to process recyclables and biodegradable waste.
4. **Establishment of Materials Recovery Facility (MRF)**
   - LGUs must establish **MRF** in every barangay to sort, recycle, and compost waste.

Key Provisions

5. Prohibition of Open Dumping and Burning

- Open dumping of waste and burning of garbage are **illegal** to prevent pollution and health hazards.

6. Solid Waste Management Planning

- LGUs must prepare and implement a **10-year solid waste management plan**, reviewed annually.

7. Public Education and Participation

- Programs to **educate citizens** on proper waste handling and encourage active participation.

8. Funding and Incentives

- LGUs can receive funds and incentives to support **sustainable waste management programs**.

Bioremediation

- **Bioremediation** is the use of living organisms—such as **bacteria, fungi, or plants** to **break down, detoxify, or remove pollutants** from contaminated **soil, water**, or **air.**

Key Types

1. **Microbial bioremediation** – bacteria degrade oil, pesticides, or sewage.
2. **Phytoremediation** – plants absorb or stabilize heavy metals and toxins.
3. **Mycoremediation** – fungi decompose complex pollutants like petroleum products.

Type of Soil Contamination	Organism Used	Bioremediation Method		Example
Oil-contaminated soil	**Bacteria** (e.g., Pseudomonas spp.)	Microbial bioremediation	Bacteria degrade petroleum hydrocarbons into less harmful substances.	Soil cleanup after oil leaks near ports and fuel depots in Manila Bay.
Heavy metal–contaminated soil	**Plants** (sunflower, vetiver grass)	Phytoremediation	Plants absorb metals like lead and cadmium, which are removed during harvesting.	Mine-affected areas in Benguet and Nueva Vizcaya.
Pesticide-contaminated soil	**Fungi** (white-rot fungi)	Mycoremediation	Fungal enzymes break down persistent pesticide residues.	Intensive vegetable farming areas in Benguet and Laguna.